

Supplementary Material

Model-Guided Design of Porous Polyimide/Boron Nitride Films for Low-Permittivity, Thermally Conductive Packaging

Jinyoung Kim

School of Chemical and Biomolecular Engineering, Georgia Institute of Technology, Atlanta, GA 30332, USA

S1 Supplementary model figures

S1.1 Shape-factor sensitivity analysis

Figure S1 examines the sensitivity of the three-phase Lewis–Nielsen predictions to the h-BN shape factor A over the range 0.3–6.0 (needle-like rods to aligned platelets), evaluated at the NP mean porosity of 65.6% with $\phi_{\max} = 0.60$ fixed. For the dielectric model (Figure S1a), all A curves are nearly indistinguishable, confirming that ε_r is dominated by the pore term and that the fitted dielectric A_{BN} is weakly determined. For the thermal model (Figure S1b), all $A = 0.3$ –6.0 curves remain below the experimental NP 70 wt% value (0.16–0.18 vs. 0.305 W m⁻¹ K⁻¹), reinforcing that the 70 wt% enhancement is driven by proximity to maximum packing and incipient particle contact rather than by any plausible single-particle shape factor. Full discussion appears in Sections 3.2 and 4.6 of the main text.

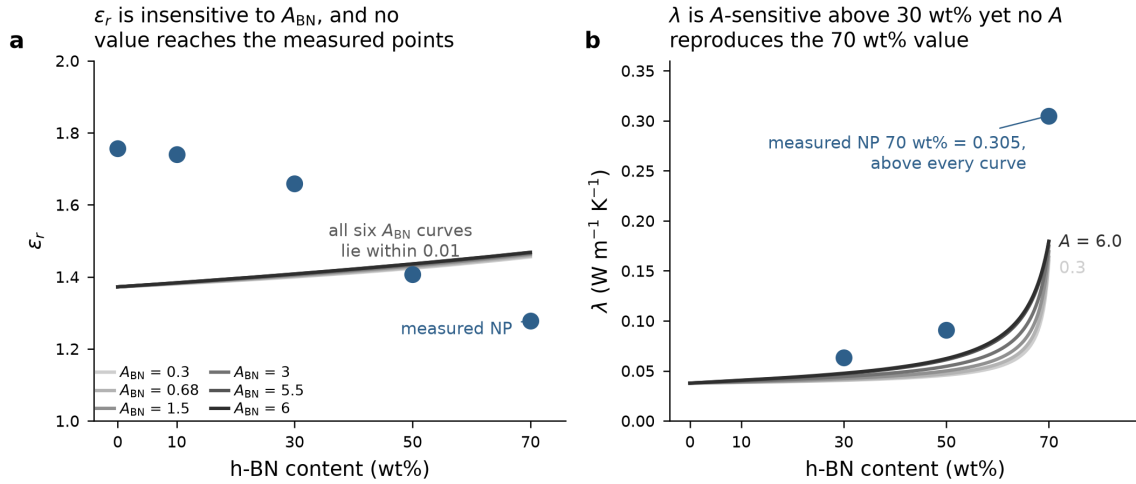


Figure S1: Sensitivity of three-phase LN predictions to h-BN shape factor A ($A = 0.3$ –6.0) for (a) dielectric constant and (b) thermal conductivity, evaluated at the NP mean porosity of 65.6% ($\phi_{\max} = 0.60$ fixed). Symbols: measured NP data. In (a) all six curves lie within 0.01 of one another, confirming that ε_r is pore-dominated and A_{BN} weakly determined; note also that no value of A brings the curves onto the measured points. In (b) the curves separate above 30 wt% yet all remain far below the measured NP 70 wt% value of 0.305 W m⁻¹ K⁻¹, so that enhancement cannot be attributed to any plausible single-particle shape factor.

S1.2 Parity plot of the dielectric model

Figure S2 compares the three-phase LN prediction with the measured dielectric constant for all ten films. The model RMSE (0.43 NP, 0.36 HP) exceeds the standard deviation of the measured data (0.19 NP, 0.18 HP), so the coefficient of determination is negative (Section 4.2 of the main text); the numerical values are listed in Table 2 of the main text.

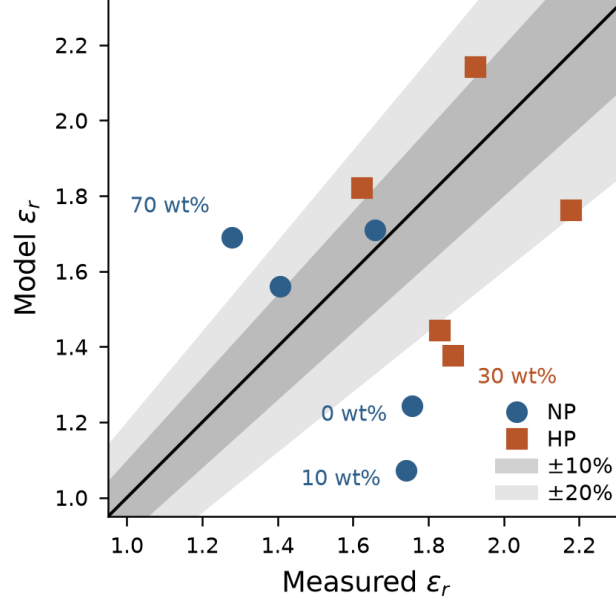


Figure S2: Parity plot of predicted vs. measured dielectric constant for all ten films (three-phase LN model evaluated at each film's open porosity), with $\pm 10\%$ and $\pm 20\%$ bands. Labels identify the compositions with the largest residuals.

S1.3 Porosity sensitivity at fixed composition

Figure S3 shows the calibrated ε_r and λ at fixed 50 wt% h-BN as porosity is varied. Both properties rise as porosity falls, so porosity alone cannot decouple them; the same coupling is shown for every composition by the loci of Figure 4c of the main text.

S1.4 Feasible processing window

Figure S4 shows the porosity range that satisfies each of the three target specifications defined in Section 4.8 of the main text (Basic: $\varepsilon_r < 2.0$, $\lambda > 0.05 \text{ W m}^{-1} \text{ K}^{-1}$; Advanced: $\varepsilon_r < 1.8$, $\lambda > 0.08 \text{ W m}^{-1} \text{ K}^{-1}$; Future: $\varepsilon_r < 1.6$, $\lambda > 0.12 \text{ W m}^{-1} \text{ K}^{-1}$), derived from the design maps of Figure 4a,b of the main text.

S1.5 Porosity that maximizes thermal conductivity under a dielectric ceiling

Figure S5 shows, for each h-BN loading, the porosity that maximizes λ under a given ε_r ceiling. The five composition curves stay within 10 percentage points of one another, so the porosity target is set almost entirely by the dielectric ceiling while the achievable λ at that porosity is set by the loading (Section 4.8 of the main text).

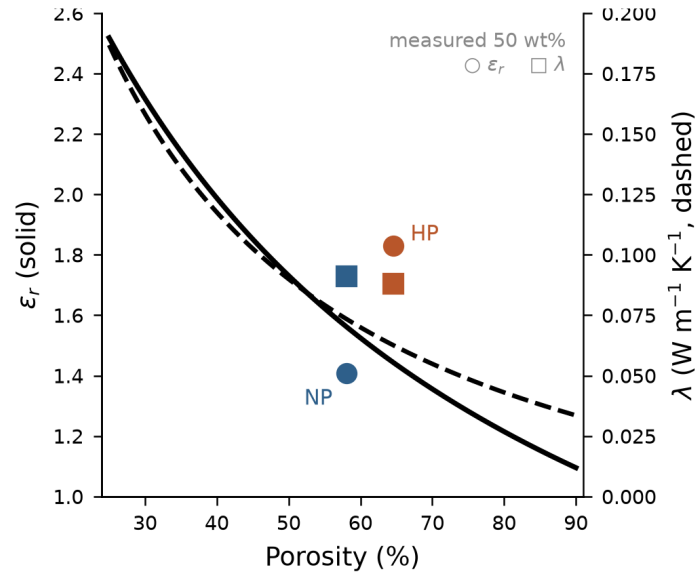


Figure S3: Porosity sensitivity at fixed 50 wt% h-BN: ϵ_r (left axis, solid) and λ (right axis, dashed) from the calibrated three-phase LN model. Circles and squares mark the measured NP and HP 50 wt% films (ϵ_r and λ , respectively).

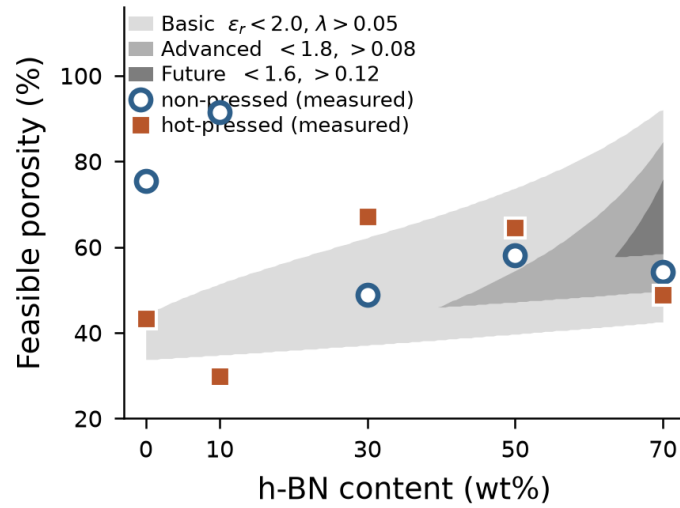


Figure S4: Feasible processing window: the porosity range meeting each target specification vs. h-BN content (nested shading, light to dark for Basic→Future); measured NP (open circles) and HP (filled squares) states overlaid.

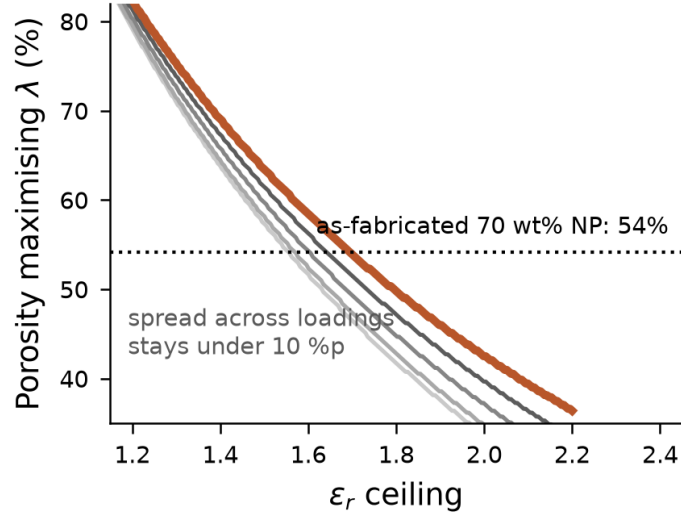


Figure S5: Porosity that maximizes λ under each ϵ_r ceiling for 0–70 wt% h-BN (light to dark grey, 0–50 wt%; orange, 70 wt%). The dotted line marks the as-fabricated porosity of the 70 wt% NP film (54%).

S2 Additional materials characterization

S2.1 Film photographs

Figure S6 shows the as-imidized (NP) and hot-pressed (HP) films. The neat and 10 wt% films are translucent and show the millimetre-scale surface texture of the NIPS sponge; from 30 wt% upward the films are opaque and matte because the h-BN platelets scatter light, and hot-pressing smooths the surface at every composition.

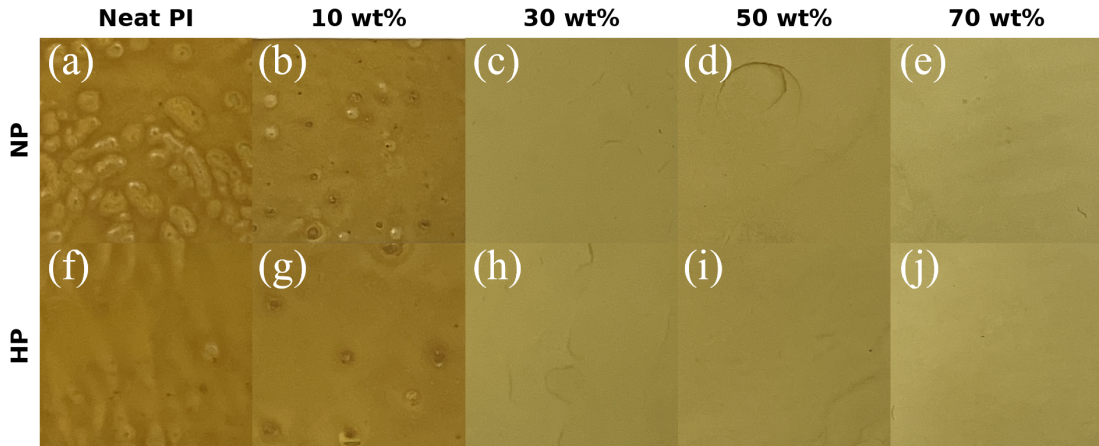


Figure S6: Photographs of the PI/h-BN films: (a–e) non-pressed and (f–j) hot-pressed at 0, 10, 30, 50, and 70 wt% h-BN (columns). Each frame shows a ~ 1 cm square region of the film surface under identical illumination.

S2.2 Top-view SEM

Figure S7 shows the film surfaces at 30, 50, and 70 wt% h-BN. At $\times 500$ the surface porosity increases with loading, from a largely closed skin at 30 wt% to an open, uniformly perforated surface at 70 wt%. At $\times 5,000$ the 30 wt% film shows agglomerates of near-spherical particles of ~ 100 – 300 nm on a smooth PI

skin, whereas the 50 and 70 wt% films show platelets embedded in and protruding from the pore walls. The change from a closed skin to an open surface with loading is consistent with the higher open (intrusion) porosity of the 70 wt% NP film relative to its total porosity (Table 3 of the main text).

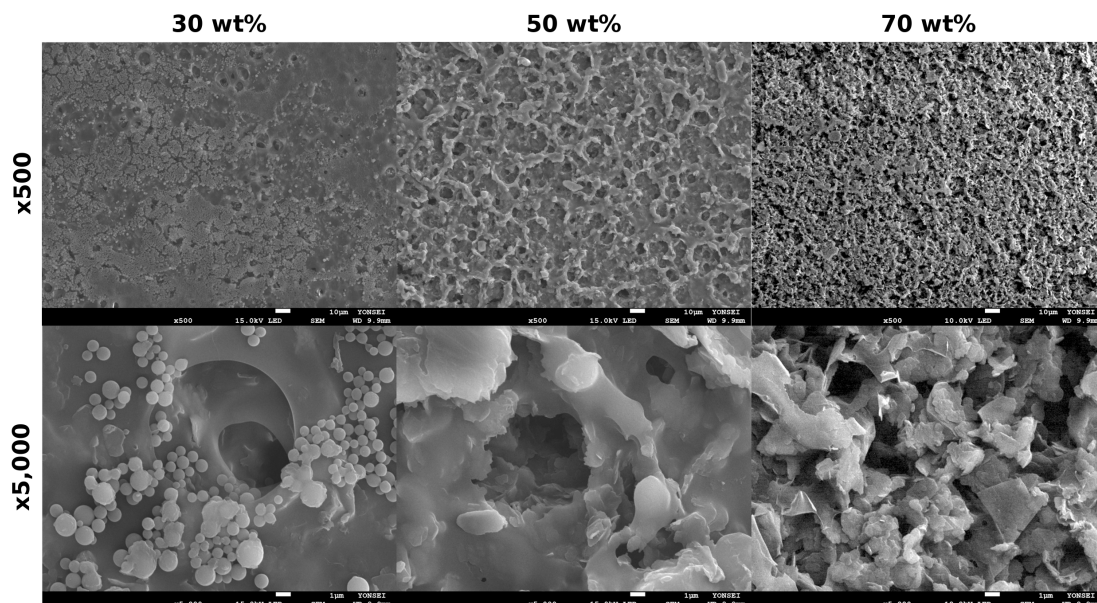


Figure S7: Top-view FE-SEM micrographs of NP PI/h-BN films at 30, 50, and 70 wt% h-BN (columns) at $\times 500$ (top row, scale bar 10 μm) and $\times 5,000$ (bottom row, scale bar 1 μm).

S2.3 FT-IR

Figure S8 confirms complete imidization at every composition. All spectra show the imide C=O asymmetric and symmetric stretches at 1778 and 1722 cm^{-1} , the aromatic C=C band at $\sim 1500 \text{ cm}^{-1}$, and the imide C–N–C stretch at 1366 cm^{-1} ; the amide carbonyl band of the poly(amic acid) precursor at $\sim 1660 \text{ cm}^{-1}$ is absent. In the composites the in-plane B–N stretch of h-BN ($\sim 1370 \text{ cm}^{-1}$) overlaps the C–N–C band, and the out-of-plane B–N bending mode at $\sim 810 \text{ cm}^{-1}$ grows monotonically with loading; both bands are absent in the neat film. The band positions quoted in Section 2.3 of the main text are those read from these spectra.

S2.4 EDS elemental analysis

S2.5 Porosity

Open porosity was taken from the total intruded volume of the mercury intrusion run on each film (PM33GT, Quantachrome), and total porosity from the bulk density (Section S3.2); the two are listed together in Table 3 of the main text and compared in Section 4.1. The intrusion curves were used for the open-porosity values only; the pore-size distributions are available from the author on request.

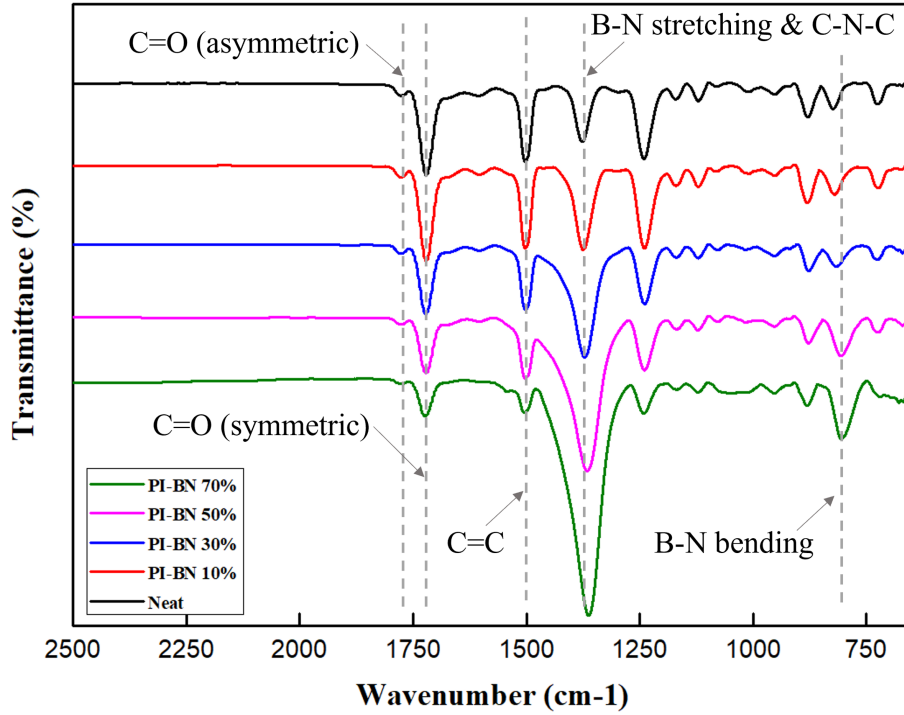


Figure S8: FT-IR transmittance spectra (Jasco FT/IR-460 Plus, 600–4000 cm^{-1} ; the 2500–650 cm^{-1} region is shown) of NP films at 0–70 wt% h-BN, offset vertically. Dashed lines mark the imide C=O stretches (1778, 1722 cm^{-1}), the aromatic C=C band ($\sim 1500 \text{ cm}^{-1}$), the overlapping C–N–C and B–N stretches (1366 cm^{-1}), and the B–N bending mode ($\sim 810 \text{ cm}^{-1}$).

Table S1: EDS elemental composition (wt%) of PI/h-BN films (FE-SEM/EDS, JEOL-7800F). B content scales with nominal h-BN loading at ≥ 10 wt% and is preserved upon hot-pressing.

Sample	B	C	N	O	Total
NP neat	5.41 [†]	73.72	3.83	17.04	100
NP 10 wt%	7.99	65.55	10.42	16.04	100
NP 30 wt%	14.70	54.28	19.36	11.66	100
NP 50 wt%	19.80	44.10	26.49	9.60	100
NP 70 wt%	29.28	26.65	38.62	5.45	100
HP neat	—	75.17	0.00	24.83	100
HP 10 wt%	8.54	68.30	8.67	14.49	100
HP 30 wt%	14.77	54.74	18.85	11.64	100
HP 50 wt%	21.11	43.12	26.62	9.15	100
HP 70 wt%	28.17	29.15	36.48	6.20	100

A second NP 70 wt% spot gave B/C/N/O = 28.81/27.93/37.01/6.25 wt%, confirming spatial uniformity. The theoretical B content of h-BN is 43.6 wt%, giving 4.4, 13.1, 21.8, and 30.5 wt% B at 10, 30, 50, and 70 wt% h-BN. [†]The apparent B signal in the neat NP film, which contains no boron, reflects the EDS quantification limit for light elements (B $K\alpha$ overlaps the C/N/O region) and is treated as background.

S3 Model details and benchmarks

S3.1 Sequential three-phase Lewis–Nielsen formulation

The Lewis–Nielsen model gives the effective scalar transport property k of a two-phase particulate composite with matrix property k_m , filler property k_f , and filler volume fraction ϕ as

$$\frac{k}{k_m} = \frac{1 + A B \phi}{1 - B \psi \phi}, \quad (\text{S1})$$

$$B = \frac{k_f/k_m - 1}{k_f/k_m + A}, \quad (\text{S2})$$

$$\psi = 1 + \frac{1 - \phi_{\max}}{\phi_{\max}^2} \phi, \quad (\text{S3})$$

where A is the Einstein coefficient (shape factor) of the dispersed phase and $\phi_{\max}$ its maximum packing fraction. Eq. S1 has three useful limits. For $A \rightarrow 0$ it reduces to the series (inverse rule-of-mixtures) model, $k^{-1} = (1 - \phi)/k_m + \phi/k_f$ when $\psi \rightarrow 1$; for $A \rightarrow \infty$ it reduces to the parallel rule of mixtures; and for $A = 1.5$, $\psi = 1$ it is the Maxwell–Garnett expression for dilute spheres. The shape factor therefore interpolates between the Wiener bounds, and a small fitted A means that the dispersed phase acts nearly in series with the matrix.

For the porous composite the model is applied twice. Step 1 mixes the PI matrix with h-BN at the solid-phase volume fraction ϕ_{BN} (Section S3.2):

$$\begin{aligned} k_{\text{dense}} &= k_{\text{PI}} \frac{1 + A_{\text{BN}} B \phi_{\text{BN}}}{1 - B \psi \phi_{\text{BN}}}, \\ B &= \frac{k_{\text{BN}}/k_{\text{PI}} - 1}{k_{\text{BN}}/k_{\text{PI}} + A_{\text{BN}}}, \quad \psi = 1 + \frac{1 - \phi_{\max, \text{BN}}}{\phi_{\max, \text{BN}}^2} \phi_{\text{BN}}. \end{aligned} \quad (\text{S4})$$

Step 2 introduces the air pores at porosity ϕ_p into the dense composite:

$$\begin{aligned} k_{\text{film}} &= k_{\text{dense}} \frac{1 + A_{\text{pore}} B_p \phi_p}{1 - B_p \psi_p \phi_p}, \\ B_p &= \frac{k_{\text{air}}/k_{\text{dense}} - 1}{k_{\text{air}}/k_{\text{dense}} + A_{\text{pore}}}, \quad \psi_p = 1 + \frac{1 - \phi_{\max, \text{pore}}}{\phi_{\max, \text{pore}}^2} \phi_p. \end{aligned} \quad (\text{S5})$$

Because $k_{\text{air}} < k_{\text{dense}}$, B_p is negative and step 2 always lowers the property. The same equations are used for the dielectric constant ($k = \varepsilon$: $\varepsilon_{\text{PI}} = 3.5$, $\varepsilon_{\text{BN}} = 6.0$, $\varepsilon_{\text{air}} = 1.0$) and for the thermal conductivity ($k = \lambda$: $\lambda_{\text{PI}} = 0.12$, $\lambda_{\text{air}} = 0.026 \text{ W m}^{-1} \text{ K}^{-1}$, $\lambda_{\text{BN}} = \lambda_{\text{BN,eff}}$ fitted).

The film-level volume fractions used by the zero-parameter mixing rules of Section S3.5 are

$$f_{\text{air}} = \phi_p, \quad f_{\text{BN}} = \phi_{\text{BN}} (1 - \phi_p), \quad f_{\text{PI}} = (1 - \phi_{\text{BN}})(1 - \phi_p). \quad (\text{S6})$$

The residual of film i in state s is

$$r_{i,s} = \varepsilon_{r,\text{model}}(\phi_{\text{BN}}[i], \phi_p[i, s]; \mathbf{p}) - \varepsilon_{r,\text{meas}}[i, s], \quad (\text{S7})$$

and the dielectric parameters $\mathbf{p} = (A_{\text{BN}}, \phi_{\text{max,BN}}, A_{\text{pore}}, \phi_{\text{max,pore}})$ are obtained by minimizing the mean squared residual over the ten films,

$$\mathcal{L}(\mathbf{p}) = \frac{1}{10} \sum_{s \in \{\text{NP}, \text{HP}\}} \sum_{i=1}^5 r_{i,s}^2, \quad \text{RMSE} = \sqrt{\mathcal{L}}, \quad (\text{S8})$$

subject to the bounds $0.05 \leq A \leq 20$ and $0.30 \leq \phi_{\text{max}} \leq 0.99$. The thermal objective has the same form over the seven films with measured λ .

S3.2 Volume-fraction conversion

The h-BN weight fraction of the solids, w , was converted to the solid-phase volume fraction with $\rho_{\text{PI}} = 1.42 \text{ g cm}^{-3}$ and $\rho_{\text{BN}} = 2.29 \text{ g cm}^{-3}$,

$$\begin{aligned} \phi_{\text{BN}} &= \frac{w/\rho_{\text{BN}}}{w/\rho_{\text{BN}} + (1-w)/\rho_{\text{PI}}}, \\ \rho_{\text{solid}} &= \left(\frac{w}{\rho_{\text{BN}}} + \frac{1-w}{\rho_{\text{PI}}} \right)^{-1}, \quad \phi_{\text{total}} = 1 - \frac{\rho}{\rho_{\text{solid}}}, \end{aligned} \quad (\text{S9})$$

where ρ is the measured bulk density (Table 3 of the main text). Table S2 lists the resulting fractions for all ten films.

Table S2: Composition and volume fractions of the ten films. ϕ_{BN} : h-BN volume fraction of the solid phase; f_{PI} , f_{BN} , f_{air} : film volume fractions from Eq. S6 using the open porosity; ϕ_{total} from Eq. S9.

h-BN (wt%)	ϕ_{BN}	$\rho_{\text{solid}} \text{ (g cm}^{-3}\text{)}$	State	ϕ_{open}	f_{PI}	f_{BN}	f_{air}	ϕ_{total}
0	0	1.420	NP	0.754	0.246	0	0.754	—
			HP	0.434	0.566	0	0.434	—
10	0.0645	1.476	NP	0.914	0.080	0.006	0.914	—
			HP	0.298	0.657	0.045	0.298	0.405
30	0.210	1.603	NP	0.488	0.405	0.107	0.488	0.798
			HP	0.672	0.259	0.069	0.672	0.716
50	0.383	1.753	NP	0.581	0.259	0.160	0.581	0.706
			HP	0.646	0.219	0.135	0.646	0.613
70	0.591	1.934	NP	0.542	0.187	0.271	0.542	0.728
			HP	0.488	0.209	0.303	0.488	0.721

S3.3 Sensitivity of the dielectric fit to the porosity definition

The main-text calibration uses the open (intrusion) porosity, which is available for all ten films. Because ε_r responds to the total air fraction, we re-fitted the dielectric model with ϕ_{total} as the porosity input for the seven films with measured density (Table S3). With the open porosity, the main-text parameter set gives RMSE 0.31 and $R^2 = -0.9$ on these seven films; with the total porosity, the re-fit gives RMSE 0.24 and $R^2 = -0.2$. The porosity definition therefore accounts for a substantial part of the residual, as stated in Section 4.2 of the main text, although the model still does not outperform the data mean. The re-fit moves the pore shape factor from 0.08 to 0.43 (with A_{BN} again weakly determined), which remains well below the isolated-sphere value of 1.5, so the qualitative conclusion that the NIPS sponge shields the field far less efficiently than spherical pores holds for either porosity definition; the numerical value of A_{pore}

quoted in the main text is specific to the open-porosity input. The sensitivity of the model to the porosity input is $\partial\varepsilon_r/\partial\phi_p \approx -1.7$ to -2.2 per unit porosity at 50–70 wt%, so an uncertainty of 3 percentage points in ϕ_p corresponds to 0.05–0.07 in ε_r , and the 21–31 percentage-point gap between open and total porosity at 30–70 wt% NP corresponds to 0.4–0.6.

Table S3: Dielectric calibration with open vs. total porosity as the input to Eq. S5. Open-porosity fit: main-text parameters, ten films; the same parameters are also evaluated on the seven films with measured density. Total-porosity fit: re-fitted on those seven films.

Porosity input	Films	A_{BN}	A_{pore}	$\phi_{\text{max,BN}}, \phi_{\text{max,pore}}$	RMSE	R^2
Open (main text)	10	0.68	0.08	0.99, 0.99	0.39 (0.43 NP, 0.36 HP)	−3.9 NP, −3.0 HP
Open (same parameters)	7	0.68	0.08	0.99, 0.99	0.31	−0.9
Total (re-fit)	7	8–12 ^a	0.43	0.99, 0.99	0.24	−0.2

^aWeakly determined: values of A_{BN} between 8 and 12 give the same RMSE to within 0.001.

S3.4 Uncertainty propagation

Thermal conductivity. With $\lambda = \alpha\rho C_p$ and independent relative uncertainties of 5% (α , laser flash), 3% (C_p , DSC) and 8% (ρ , dominated by the thickness measurement of a porous film), the propagated relative uncertainty is $\sqrt{0.05^2 + 0.03^2 + 0.08^2} = 9.9\%$, i.e. about $\pm 0.03 \text{ W m}^{-1} \text{ K}^{-1}$ at 70 wt% and ± 0.005 – $0.01 \text{ W m}^{-1} \text{ K}^{-1}$ at 10–50 wt%. The +16% residual of the NP 70 wt% film and the −8.5% change on hot-pressing (Section 4.3) are therefore comparable to, not larger than, the measurement uncertainty, and are interpreted as indicative only.

Dielectric constant. The repeatability of the in-house LCR measurement is about $\pm[0.05]$ ($\pm 3\%$). The inter-laboratory comparison on the HP 30 wt% film (1.864 in-house vs. 1.610 at KOPTRI, −13.6%) is a better estimate of the reproducibility for porous films with contact electrodes, because it includes the electrode–surface air gap discussed in Section 4.2. We therefore regard model residuals below $\sim 14\%$ as within the experimental reproducibility; residuals of 19–38% (Table 2 of the main text) exceed it.

Porosity. The repeatability of mercury intrusion is about ± 2 percentage points. The density-derived total porosity inherits the 8% uncertainty of ρ : $\delta\phi_{\text{total}} = 0.08 \rho/\rho_{\text{solid}} = 2$ –5 percentage points (largest for the densest film, HP 10 wt%). Through the sensitivity quoted in Section S3.3, both correspond to ± 0.04 –0.1 in the model ε_r .

CTE. The TMA repeatability is about $\pm[1]$ ppm K^{−1}, small compared with the 15–30 ppm K^{−1} separation between the measured values and the Turner and rule-of-mixtures bounds (Table 4 of the main text).

S3.5 Benchmark against standard mixing rules

Table S4 compares the calibrated three-phase LN model with the zero-parameter three-component mixing rules evaluated with the film volume fractions of Eq. S6: the Lichtenecker logarithmic rule, $\ln \varepsilon = \sum_j f_j \ln \varepsilon_j$; the Maxwell–Garnett rule with PI as the host and h-BN and air as inclusions; the symmetric Bruggeman rule, $\sum_j f_j (\varepsilon_j - \varepsilon)/(\varepsilon_j + 2\varepsilon) = 0$; and, as bounds, the parallel (Wiener upper) and series (Wiener lower) rules. Over the 30–70 wt% NP films, the range in which the model is used for design, the calibrated LN model gives RMSE 0.26 against 0.55 (Lichtenecker), 0.73 (Bruggeman) and 0.85 (Maxwell–Garnett);

over all ten films the corresponding values are 0.39 against 0.48, 0.58 and 0.64. All three symmetric or host-based rules overpredict ε_r because they treat the pores as inclusions that only partly interrupt the field. The series bound, which corresponds to the $A \rightarrow 0$ limit of Eq. S1, lies closest to the NP data at 30–70 wt% (RMSE 0.17); the fitted $A_{\text{pore}} = 0.08$ places the pore step of the LN model close to this limit, which is the quantitative expression of the statement in Section 3.2 of the main text that the bicontinuous sponge pores act nearly in series with the PI skeleton.

Table S4: RMSE of the dielectric constant for the calibrated LN model and zero-parameter mixing rules ($\varepsilon_{\text{PI}} = 3.5$, $\varepsilon_{\text{BN}} = 6.0$, $\varepsilon_{\text{air}} = 1$; film volume fractions from the open porosity). “30–70 wt%” includes the six NP and HP films of that range; “30–70 NP” the three NP films quoted in Section 3.1 of the main text.

Model	Parameters	All ten films	30–70 wt%	30–70 wt% NP
Three-phase LN (calibrated)	2–3 active	0.39	0.32	0.26
Lichtenecker	0	0.48	0.48	0.55
Bruggeman (symmetric)	0	0.58	0.63	0.73
Maxwell–Garnett (PI host)	0	0.64	0.72	0.85
Parallel (Wiener upper bound)	0	0.88	1.04	1.19
Series (Wiener lower bound)	0	0.40	0.32	0.17

S4 Parameter estimation protocol

S4.1 Dielectric calibration

The objective of Eq. S8 was minimized with the Nelder–Mead simplex algorithm (SciPy `optimize.minimize`; $x_{\text{tol}} = 10^{-6}$, $f_{\text{tol}} = 10^{-9}$, at most 4000 iterations) from 16 random starting points drawn uniformly from $A_{\text{BN}} \in [0.3, 6]$, $A_{\text{pore}} \in [0.02, 2]$ and $\phi_{\text{max}} \in [0.5, 0.99]$ for both phases; parameter sets outside the bounds of Section S3.1 were penalized. All starts converged to the same ridge: both packing fractions run to the upper bound of 0.99, A_{pore} lies at 0.07–0.09, and the objective is nearly flat in A_{BN} (Table S5). Over $A_{\text{BN}} = 0.05$ –6 the RMSE varies by only 0.003, so the dielectric data determine A_{pore} but not A_{BN} ; the value $A_{\text{BN}} = 0.68$ quoted in the main text lies on this ridge and is reported with that qualification. Fixing $\phi_{\text{max,BN}}$ anywhere between 0.50 and 0.99 and re-fitting the remaining parameters changes the RMSE by less than 0.01 (Table S6), which is why the packing fractions are described as inactive. With the main-text parameter set the model reproduces the NP and HP data with RMSE 0.43 and 0.36, against standard deviations of the measured values of 0.19 and 0.18, so $R^2 = -3.9$ (NP) and -3.0 (HP).

Table S5: Profile of the dielectric objective along A_{BN} : for each fixed A_{BN} , the remaining parameters were re-optimized (multi-start Nelder–Mead).

A_{BN}	0.05	0.20	0.68	1.5	6.0
A_{pore}	0.087	0.084	0.078	0.073	0.066
$\phi_{\text{max,BN}}, \phi_{\text{max,pore}}$	0.99, 0.99	0.99, 0.99	0.99, 0.99	0.99, 0.99	0.99, 0.99
RMSE (ten films)	0.393	0.393	0.394	0.395	0.396

S4.2 Thermal calibration and the degeneracy of the unconstrained fit

The thermal model has three step-1 parameters ($\lambda_{\text{BN,eff}}, A_{\text{BN},\lambda}, \phi_{\text{max,BN},\lambda}$); the pore step uses the dielectric pore parameters ($A_{\text{pore}} = 0.08$, $\phi_{\text{max,pore}} = 0.99$), since the pore geometry is a property of the film, not

Table S6: Sensitivity of the dielectric calibration to the h-BN packing fraction: $\phi_{\max, \text{BN}}$ fixed at the stated value, A_{BN} , A_{pore} and $\phi_{\max, \text{pore}}$ re-fitted.

$\phi_{\max, \text{BN}}$	0.50	0.60	0.70	0.80	0.90	0.99
A_{pore}	0.056	0.059	0.061	0.062	0.079	0.087
RMSE (ten films)	0.398	0.397	0.397	0.397	0.394	0.393

of the transported quantity. With $A_{\text{BN}, \lambda} = 5.5$ and $\phi_{\max, \text{BN}, \lambda} = 0.60$ fixed, the single-parameter fit to the seven measured conductivities gives $\lambda_{\text{BN}, \text{eff}} = 6.65 \text{ W m}^{-1} \text{ K}^{-1}$ with $\lambda_{\text{air}} = 0.026 \text{ W m}^{-1} \text{ K}^{-1}$ (6.81 with $\lambda_{\text{air}} = 0.023$; the predicted film conductivities differ by less than $0.001 \text{ W m}^{-1} \text{ K}^{-1}$ between the two), RMSE 0.027 (NP) and $0.030 \text{ W m}^{-1} \text{ K}^{-1}$ (HP), and predictions of 0.262 and $0.315 \text{ W m}^{-1} \text{ K}^{-1}$ for the NP and HP 70 wt% films.

When all three parameters are released, the multi-start optimizer runs to the edge of the admissible region: $A_{\text{BN}, \lambda}$ increases without bound (it reaches the upper limit of 2000 imposed in the code), $\phi_{\max, \text{BN}, \lambda}$ falls to 0.06, $\lambda_{\text{BN}, \text{eff}}$ falls to $1.9 \text{ W m}^{-1} \text{ K}^{-1}$, and the RMSE improves only from 0.027/0.030 to 0.025 (pooled). Table S7 shows why: for any fixed $A_{\text{BN}, \lambda}$ between 20 and 700 there is a compensating $\phi_{\max, \text{BN}, \lambda}$ that gives the same pooled RMSE to within 0.0002, because in Eq. S4 a large A and a small $\phi_{\max}$ both amplify the filler contribution near the 70 wt% point and the seven data cannot separate them. The unconstrained solution ($A \sim 10^3$, $\phi_{\max} \sim 0.1$) has no physical meaning for a platelet filler, whereas the constrained solution uses literature values for randomly oriented platelets and reproduces the data equally well. This is the basis for the single-parameter calibration adopted in the main text.

Table S7: Profile of the thermal objective along $A_{\text{BN}, \lambda}$: for each fixed A , $\lambda_{\text{BN}, \text{eff}}$ and $\phi_{\max, \text{BN}, \lambda}$ were re-optimized (pooled RMSE over the seven measured films, $\text{W m}^{-1} \text{ K}^{-1}$). The constrained calibration of the main text is the $A = 5.5$, $\phi_{\max} = 0.60$ row.

$A_{\text{BN}, \lambda}$	$\phi_{\max, \text{BN}, \lambda}$	$\lambda_{\text{BN}, \text{eff}}$ ($\text{W m}^{-1} \text{ K}^{-1}$)	Pooled RMSE	Comment
5.5 (fixed)	0.60 (fixed)	6.65	0.028	Main-text calibration
20	0.53	4.2	0.025	
50	0.34	2.4	0.025	
100	0.25	2.1	0.025	
300	0.15	2.0	0.025	
700	0.10	1.9	0.025	Unconstrained optimum region

S5 Python implementation

The listing below (also provided as `code/ln3_model.py`) implements Eqs. S1–S9, the multi-start calibration of Section S4.1, the total-porosity re-fit of Section S3.3, and the thermal fits of Section S4.2. It requires NumPy and SciPy only. The design maps and loci of Figure 4 of the main text and Figures S3–S5 are obtained by evaluating `ln3` on a grid of (h-BN content, porosity). **[Deposit the code in a public repository and reference the DOI here and in the Data availability statement.]**

```

"""Three-phase Lewis-Nielsen (LN) model for porous PI/h-BN films.

Reproduces the calibration, benchmarks and sensitivity analyses of the
Supplementary Material (Sections S3-S4). Requires numpy and scipy.
"""
import numpy as np

```

```

from scipy.optimize import minimize

# ----- data
w = np.array([0, 10, 30, 50, 70]) / 100.0          # h-BN weight fraction of solids
RHO_PI, RHO_BN = 1.42, 2.29                        # g cm-3
phi_BN = (w / RHO_BN) / (w / RHO_BN + (1 - w) / RHO_PI)  # solid-phase volume fraction
rho_solid = 1 / (w / RHO_BN + (1 - w) / RHO_PI)
eps_meas = {"NP": np.array([1.757, 1.740, 1.659, 1.407, 1.278]),
            "HP": np.array([2.178, 1.924, 1.864, 1.829, 1.622])}
lam_meas = {"NP": np.array([np.nan, np.nan, 0.063, 0.091, 0.305]),
            "HP": np.array([np.nan, 0.114, 0.051, 0.088, 0.279])}
phi_open = {"NP": np.array([75.4, 91.4, 48.8, 58.1, 54.2]) / 100,
            "HP": np.array([43.4, 29.8, 67.2, 64.6, 48.8]) / 100}
rho_bulk = {"NP": np.array([np.nan, np.nan, 0.323, 0.515, 0.526]),
            "HP": np.array([np.nan, 0.879, 0.455, 0.678, 0.539])}
phi_total = {s: 1 - rho_bulk[s] / rho_solid for s in rho_bulk}
EPS_PI, EPS_BN, EPS_AIR = 3.5, 6.0, 1.0
LAM_PI, LAM_AIR = 0.12, 0.026                      # W m-1 K-1

# ----- model
def ln(k_m, k_f, phi, A, phi_max):
    """Two-phase Lewis-Nielsen expression (Eqs. S1-S3)."""
    B = (k_f / k_m - 1) / (k_f / k_m + A)
    psi = 1 + (1 - phi_max) / phi_max**2 * phi
    return k_m * (1 + A * B * phi) / (1 - B * psi * phi)

def ln3(k_m, k_f, k_pore, phi_f, phi_p, A_f, pmax_f, A_p, pmax_p):
    """Sequential three-phase model: step 1 (filler), step 2 (pores)."""
    k_dense = ln(k_m, k_f, phi_f, A_f, pmax_f)
    return ln(k_dense, k_pore, phi_p, A_p, pmax_p)

def eps_model(p, phi_in=phi_open):
    A_BN, pmax_BN, A_p, pmax_p = p
    return {s: ln3(EPS_PI, EPS_BN, EPS_AIR, phi_BN, phi_in[s],
                  A_BN, pmax_BN, A_p, pmax_p) for s in ("NP", "HP")}

def lam_model(lam_BN, A=5.5, pmax=0.60, A_p=0.08, pmax_p=0.99):
    return {s: ln3(LAM_PI, lam_BN, LAM_AIR, phi_BN, phi_open[s],
                  A, pmax, A_p, pmax_p) for s in ("NP", "HP")}

def rmse(meas, pred):
    m = ~np.isnan(meas) & ~np.isnan(pred)
    return np.sqrt(np.mean((meas[m] - pred[m])**2))

def r2(meas, pred):
    m = ~np.isnan(meas) & ~np.isnan(pred)
    return 1 - np.sum((meas[m] - pred[m])**2) / np.sum((meas[m] - meas[m].mean())**2)

# ----- objective (Eq. S8)
BOUNDS = [(0.05, 20), (0.30, 0.99), (0.01, 20), (0.30, 0.99)]

def objective(p, phi_in=phi_open):
    if any(not (lo <= v <= hi) for v, (lo, hi) in zip(p, BOUNDS)):
        return 1e6
    pred = eps_model(p, phi_in)
    res = np.concatenate([(pred[s] - eps_meas[s])[~np.isnan(phi_in[s])] for s in ("NP", "HP")])
    return np.mean(res**2)

def multistart(fun, sampler, n=16, seed=0):

```

```

rng = np.random.default_rng(seed)
best = None
for _ in range(n):
    r = minimize(fun, sampler(rng), method="Nelder-Mead",
                 options={"xatol": 1e-6, "fatol": 1e-9, "maxiter": 4000})
    if best is None or r.fun < best.fun:
        best = r
return best

if __name__ == "__main__":
    # dielectric calibration with open porosity (Section S4.1)
    samp = lambda g: [g.uniform(0.3, 6), g.uniform(0.5, 0.99), g.uniform(0.02, 2), g.uniform(0.5, 0.99)]
    fit = multistart(objective, samp)
    print("dielectric fit (open porosity):", fit.x.round(3), "RMSE", np.sqrt(fit.fun).round(3))
    # published parameter set
    pub = (0.68, 0.99, 0.08, 0.99)
    pred = eps_model(pub)
    for s in ("NP", "HP"):
        print(s, pred[s].round(3), "RMSE", rmse(eps_meas[s], pred[s]).round(3), "R2", r2(eps_meas[s], pred[s]).round(2))
    # re-fit with total porosity (Section S3.3)
    fit_t = multistart(lambda p: objective(p, phi_total), samp)
    print("dielectric fit (total porosity):", fit_t.x.round(3), "RMSE", np.sqrt(fit_t.fun).round(3))
    # thermal single-parameter fit (Section S4.2)
    def obj_l(x, A=5.5, pmax=0.60):
        if x[0] <= 0 or not (0.05 < pmax < 0.99):
            return 1e6
        pr = lam_model(x[0], A, pmax)
        res = np.concatenate([(pr[s] - lam_meas[s])[~np.isnan(lam_meas[s])] for s in ("NP", "HP")])
        return np.mean(res**2)
    r = minimize(obj_l, [5.0], method="Nelder-Mead")
    pl = lam_model(r.x[0])
    print("lam_BN,eff =", r.x[0].round(3), {s: rmse(lam_meas[s], pl[s]).round(4) for s in ("NP", "HP")})
    # unconstrained three-parameter thermal fit
    fit_3 = multistart(lambda x: obj_l([x[0]], A=x[1], pmax=x[2]) if 0.05 < x[1] < 2000 else 1e6,
                      lambda g: [g.uniform(1, 50), g.uniform(0.5, 50), g.uniform(0.2, 0.95)], n=20)
    print("unconstrained thermal fit:", fit_3.x.round(3), "RMSE", np.sqrt(fit_3.fun).round(4))

```